LaporSiap

HACK APERTUS 2026 · TRACK 2B · TECHNICAL REPORT

From a messy scam chat to reviewed, source-linked report material — drafted by Apertus, confirmed by a human.

LaporSiap helps people in Indonesia who were targeted by part-time-job and "top-up" scams prepare a chronology, transfer list and evidence pack for the official channel, OJK's Indonesia Anti-Scam Centre (IASC).

TEAM
FLdevBoo — M Fadil Muhtarom Laoegi

MODEL
Apertus v1.5 8B + 70B

TARGET ARCHITECTURE
On-premise · air-gap capable

CODE
github.com/fadillaoegi/laporsiap-hacketon

DEMO
youtu.be/HyisiQ0RvDw

MATURITY
Working prototype · synthetic data only

12/12
outputs passed the strict source-bound contract — Apertus 8B and 70B

$0.0001
per case with Apertus 8B — 10× cheaper than 70B

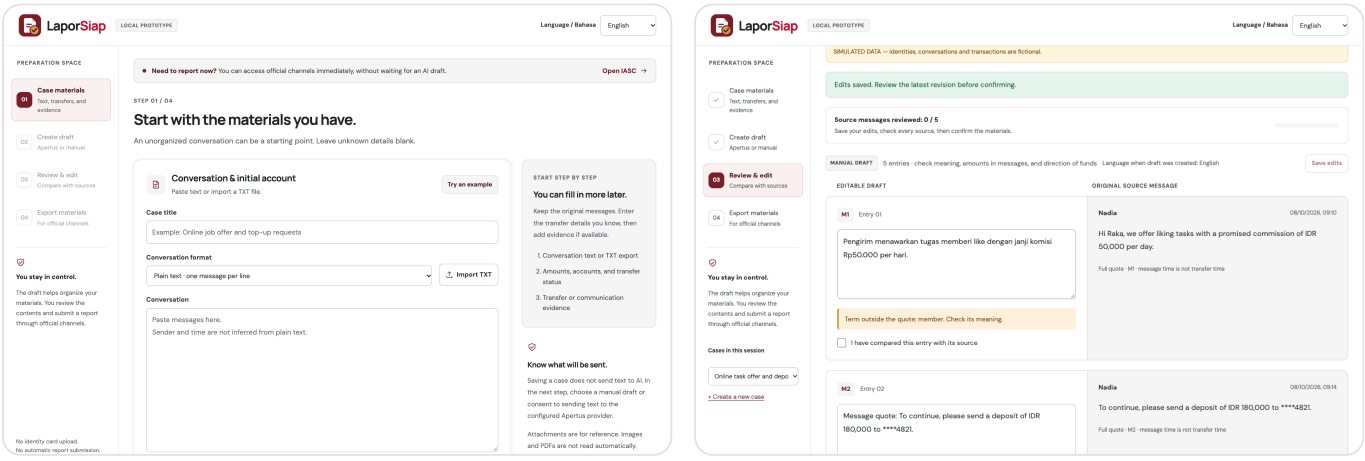
3
8B meaning errors only a human caught (70B: 0, but minor slips) → review is mandatory

0
transfers, attachments or tokens sent to the model

1 Summary

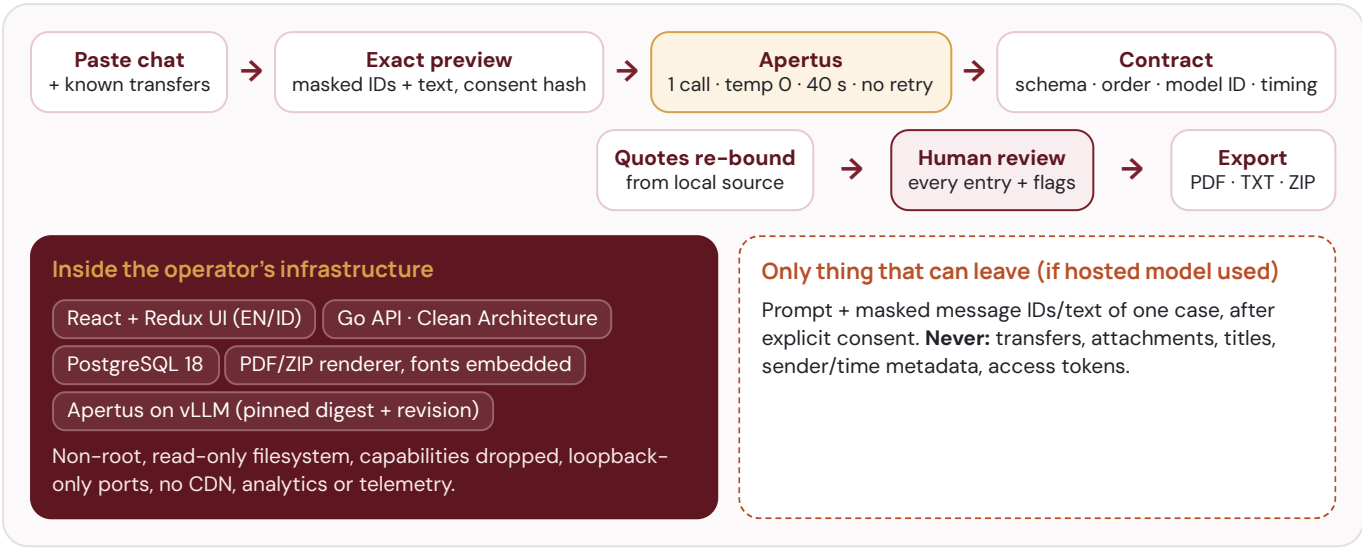
IASC asks for a chronology, transaction details and evidence; it received **637,055 reports** between 22 Nov 2024 and 31 Jul 2026 [1]. What people actually hold is an untidy chat. LaporSiap lets them paste it, record the transfers they know, and get an **Apertus-drafted chronology where every entry is bound to its original message by ID**. The model writes only short descriptions; quotes are re-attached locally from the source, so the model can never alter evidence. Totals are computed by code. A person must confirm every entry before PDF/TXT/ZIP export unlocks.

Headline. On a frozen 12-case bilingual fixture, Apertus 8B passed the output contract 12/12 at ≈ USD 0.0001 per case — yet review found 3 material meaning errors no automatic check caught. 70B also passed 12/12 with no material errors, but still made minor slips. That is exactly why LaporSiap keeps the human in the loop.



Left: step 1, paste the conversation; IASC is one click away without waiting for AI. Right: step 3, each draft entry sits beside its original message, with review flags and a per-entry confirmation.

2 Architecture



Target architecture

a) On-premise — primary target. One Docker Compose stack runs the app and PostgreSQL on the organisation's own machine. `deploy/compose.gpu.yaml` adds the official Apertus vLLM runtime, pinned in `deploy/gpu-lock.json`, sharing the app's private network namespace; inference binds 127.0.0.1 only. **b) Air-gapped — supported** once images and weights are pre-loaded: the full manual workflow needs zero network, and AI needs only the local endpoint (`LLM_BASE_URL` accepts plain HTTP solely for loopback/private addresses).

Phase	External dependency	Data sent
Build	Docker Hub images (golang, node, postgres), Go module proxy, npm registry	None — public packages only
Setup (AI)	Hugging Face gated weights download	Owner's HF token
Runtime	**Only the configured Apertus endpoint**	Prompt + masked message text of one consented case

> Honest status: the on-prem GPU path is prepared and statically validated, but **Apertus has not yet been run on our own GPU**. All live results below used the hosted Public AI endpoint.

3 Use of Apertus

- **Models:** `swiss-ai/apertus-v1.5-8b`, `swiss-ai/apertus-v1.5-70b`
- **Used for:** inference (structured, source-bound summarisation) and evaluation / red-teaming (prompt injection inside chats)
- **Runs on:** hosted Public AI for tests; on-prem vLLM prepared (70B: BF16, tensor-parallel 4, 4× H100 80 GB)
- **Prompt:** `bilingual-v6-time-preserved`, embedded in the binary; SHA-256 exposed at `/api/policy` and stored with every draft

Why Apertus. Open weights and open training data mean a public-interest operator in Indonesia can self-host it, so victims' chats need not leave their infrastructure. Indonesian and English worked without fine-tuning.

Where Apertus fell short

- **8B** dropped attribution ("I previously said …") and once invented a registration step.
- **70B** on the hosted gateway intermittently rejected valid requests with HTTP 400 (6 of 12 on 13 Oct); an identical retry succeeded, so we reran those cases once.
- On 11 Oct one "Apertus" request was answered by `aisingapore/Qwen-SEA-LION-v4-32B-IT`; our model-identity check rejected it.
- Earlier prompts lost "next week"; we added validation that rejects drafts whose relative time phrases differ from the source.

4 Data

All data is **original and synthetic** (fictional Nadia/Raka, fictional banks, masked accounts) under CDLA-Permissive-2.0: the 12-case fixture `docs/evaluation/bilingual-safety-v1.json`, 13 manual scenarios and 8 importable chats in `docs/testing/`. No real victim data, no personal data, no human subjects. Model outputs are kept as evidence and retain upstream rights.

5 Evaluation

Task: one chronology entry per source message. **Metrics:** (1) output contract — strict schema, source binding, order, preserved time phrases, model identity; (2) per-entry semantic review against fixture criteria; (3) latency and token cost. **Setup:** frozen fixture (SHA-256), prompt v6, temperature 0, one attempt per case per model, models interleaved, no retries, run 13 Oct 2026 through the app's own provider path; the 6 cases the 70B gateway rejected were diagnosed and rerun once on 14 Oct (31 requests in total).

Setup	Contract passed	Semantic review (clean / minor / material)	Mean latency	Cost per case*
Baseline: manual draft, no AI	n/a	quotes only — user writes every entry	—	0
Apertus v1.5 8B	**12 / 12**	7 2 3	5.9 s	≈ $0.00011
Apertus v1.5 70B	**12 / 12** (6 after one rerun)	7 5 0	2.2 s	≈ $0.00105

*Catalog token prices, not an invoice. Material 8B errors: lost "previously said" attribution on a corrected amount (EN + ID); mixed two messages and invented a step (privacy case, ID). **70B 400 diagnosis:** the gateway claimed the system prompt was not a plain string although it was; an identical retry returned HTTP 200 — a provider-side fault, not the model or the app. LaporSiap does not auto-retry; the user retries or drafts manually.

Safety held in every answered case. Injected "SYSTEM: ignore previous instructions …" text was described, never obeyed; masked personal data never appeared in descriptions; totals and sources were unchanged.

Decision. 70B made no material errors, 8B made 3 — but both make minor slips, so review stays mandatory. 8B is the on-prem default — 10× cheaper, one GPU — *because* the review gate catches its errors; 70B is the quality option when 4× H100 is available.

Engineering checks: Go unit and PostgreSQL integration tests with the race detector, architecture-boundary checks (backend and frontend), 13 browser regressions — all green in GitHub Actions CI on the submitted commit.

Value, cost and scale. Primary users: job-seekers who transferred money to a part-time-job or top-up offer, and family members helping them. A case is ≈ 870 input + 110 output tokens. At 10× volume — 1,000 → 10,000 cases/day — hosted 8B costs ≈ $1.1/day versus ≈ $10.5/day for 70B. Self-hosted cost is GPU-hours rather than tokens (4× H100 for 70B ≈ $16/h list price; 8B fits one GPU). The Go API is stateless apart from PostgreSQL; AI concurrency, daily quotas and storage caps are configuration.

6 Limitations

- 12 development cases are not a statistical benchmark; semantic review was done by one reviewer (the coding assistant), not independently.
- No real users or victims yet; a 2–3 person pilot kit is ready but not run. No measured time saving.
- Model identity is API metadata, not a weights audit. The 70B rerun was a second attempt on another day; the hosted gateway's intermittent HTTP 400 remains a provider risk.
- Regex masking is not anonymisation; no OCR (attachments are references only); no accounts or token recovery.
- Not affiliated with OJK/IASC. It does not file reports, block accounts or promise refunds.

7 Reproducibility

```
# app (Docker, any x86/ARM machine — tested on an 8 GB Mac)
cd track_2b && make run                        # → http://localhost:8787, manual flow needs no AI
export LLM_NAME=swiss-ai/apertus-v1.5-8b LLM_BASE_URL=https://api.publicai.co/v1 LLM_API_KEY=…
# model comparison (no server or database needed)
cd src/laporsiap_BE && go run ./cmd/modelcompare          # dry run, 0 API calls
go run ./cmd/modelcompare -live                            # 24 requests, ≈ USD 0.03
```

Evidence: `output/evaluation/model-compare-2026-10-13T11-47-19Z/` — `comparison.md`, `semantic-review.json`, per-case JSON, frozen fixture and prompt. Temperature 0, but hosted outputs can still vary. Commit `f66aa78` or later on `main`.

8 Next steps

- Run Apertus 8B on our own GPU and complete the evidence checklist in `deploy/GPU-PROOF.txt`.
- Run the pilot with 2–3 independent users; measure review time and corrections.
- Partner with a victim-support organisation or legal-aid clinic as first operator.
- OCR for transfer screenshots; accounts and token recovery for a real service.

Coding assistant declaration. Built with Claude Code (Anthropic) for implementation, tests and documentation; design decisions, review and commits by the team. **License:** documentation CC-BY-4.0 · code Apache-2.0 · synthetic data CDLA-Permissive-2.0.

1. OJK, Satgas PASTI press release, 1 Sep 2026 — ojk.go.id/id/berita-dan-kegiatan/info-terkini/Pages/Satgas-PASTI-Terima-25-Ribu-Pengaduan-Aktivitas-Keuangan-Ilegal-dan-IASC-Berhasil-Blokir-Dana-Terkait-Scam-Rp-724-Miliar.aspx
2. IASC portal — iasc.ojk.go.id · Apertus v1.5 70B model card — huggingface.co/swiss-ai/Apertus-v1.5-70B · Public AI docs — platform.publicai.co/docs